

Richard Álvarez

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Professional Summary

Applied AI systems engineer who turns frontier models into reliable tools for real business workflows. Designs agent systems, MCP and tool integrations, orchestration, identity and permission boundaries, human approval paths, observability, and cost-aware deployment across production software stacks.

Experience

AI Solutions Engineer Sep 2025 – Present
Layer 7 Systems | Independent Contractor Remote

- Built and operated a stateful TypeScript/Bun/Express MCP service connecting Retell voice agents with Autotask and Microsoft Graph/Teams for structured service-desk intake, ticket actions, and technician routing.
- Engineered caller and company resolution, contact-safe ticket access, backend authorization gates, deterministic transfer eligibility, post-call workflows, and human fallback to prevent unsafe writes and cross-client data exposure.
- Developed an authenticated React operations dashboard and supported a 750+ test regression suite with Zod validation, Pino logging, cost tracking, health checks, alerts, and failure-recovery coverage.
- Deployed Docker workloads through GitHub Actions across staging and production; diagnosed duplicate tickets, stale identity state, API drift, retry/DLQ failures, transfer defects, and CI/VPS incidents.
- Built a workflow-analysis pipeline that reviewed 60 recordings from 14 users and identified 176 AI-assistable moments; delivered a Next.js/Neon training platform with four courses and 16 modules for 34 trainees.

Freelance Web Developer Oct 2024 – Present
Richard Álvarez | Sole Proprietorship Chicago, IL

- Design and ship responsive web applications using Next.js, React, Node.js, Tailwind CSS, Stripe, PostgreSQL, and object storage for creative and commercial clients.
- Translate client requirements into scoped implementations, deployment plans, documentation, and training; maintain production hosting and content workflows after launch.

IT Assistant Sep 2023 – Feb 2024
Kenyon College Library & Information Services Gambier, OH

- Supported workstation imaging, hardware deployment, printer/AV troubleshooting, and secure drive destruction; automated classroom maintenance scheduling with a custom spreadsheet tool.

Selected Project

Vigil | TypeScript, Bun, Hono, SQLite FTS5, Next.js, Cloudflare | GitHub 2026

- Built an open-source, self-hosted email intelligence agent with multi-model routing, encrypted BYOK credentials, persistent memory, deterministic action mapping, alerts, audit logs, and per-invocation cost tracking.

Education

Kenyon College Aug 2020 – May 2024
Bachelor of Arts in Film Gambier, OH
Minor in History; Integrated Program in Humane Studies concentration

Courses: AI for the Humanities, Software Development, Data Structures & Programming Design, Senior Research Seminar, Advanced Post-Production, Elements & Theory of Post-Production

Technical Skills

Agent Systems: OpenClaw, Claude Code, Codex, GitHub Copilot/VS Code, MCP, tool calling, RAG, structured outputs, context and memory design, multi-agent orchestration
Workflow & Integration: n8n, REST APIs, webhooks, Autotask, Microsoft Graph/Teams, identity and permissions, human approval paths
Reliability & Delivery: Docker, GitHub Actions, Linux/VPS, CI/CD, automated testing, validation, logging, observability, incident response, cost monitoring
Product & Data: TypeScript, Python, Bun/Node.js, Hono/FastAPI, React/Next.js, PostgreSQL, SQLite FTS5
Systems & Media: C++20, CMake, FFmpeg, PySceneDetect; Rust under active study